Scope resolution operator ::

Scope resolution is use access the global variable in the presence of local variable with same name.

```
int data=10;
int main()
{       int data=45;
        cout<<data<<endl;
        cout<<::data<<endl;
}
```

accessing local variable

accessing global variable

endl is line feed operator

cout<<"\n" ; can be written as cout<<endl;

typeid gives the datatype of the variable

new is an operator which is used to allocate the memory dynamical
delete is an operator which is used to deallocate the dynamically allocated memory.

syntax :

```
pointer =new datatype;
 ex: int *p=new int;
     char *c;
          c=new char;
struct St *ptr=new struct St;
```

// creates integer dynamically and returns its base address

syntax of delete

```
        delete pointer;
        ex : delete p;
             delete c;
```

```
int *p=malloc(sizeof(int));
char *c=malloc(sizeof(char));
struct St *ptr=malloc(sizeof(struct St));

int *p=calloc(1,sizeof(int)));
```

syntax to allocate and initialise the memory in a single statement.

pointer=new datatype(value);

ex : int *ptr=new int(123);

syntax to create an array dynamically using new operator

pointer=new datatype[size];

ex :

```
int *p=new int[5]; // array of 5 integers
char *c=new char[10] ;// array of 10 characte
```

syntax for deallocation

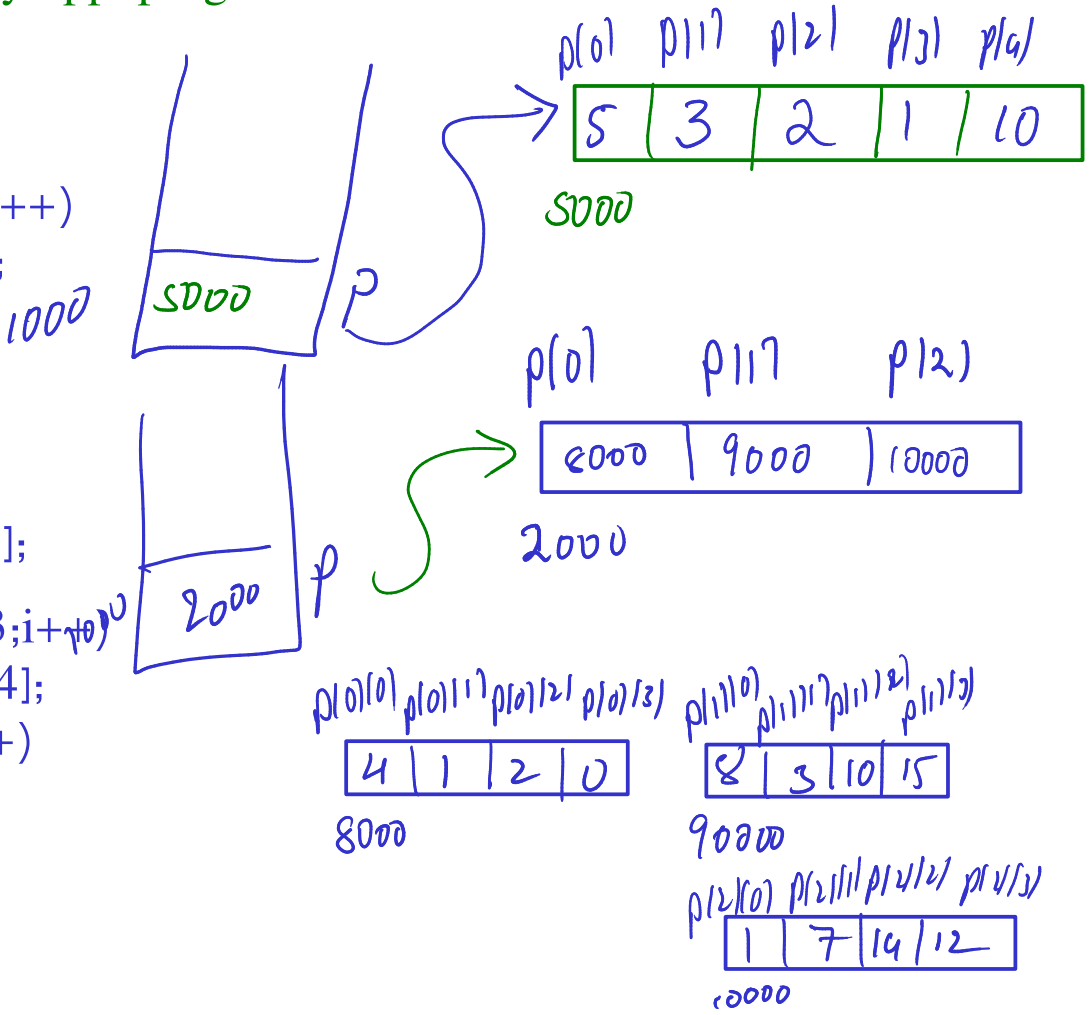
delete[]pointer;

ex : delete[]p;

delete[]c;

refer to 1darray.cpp program

```
p=new int[5];
for(int i=0;i<5;i++)
p[i]=rand()%100;
```




```
p=new int *[3];
for(int i=0;i<3;i++)
{p[i]=new int[4];
for(j=0;j<4;j++)
cin>>p[i][j];
}
```